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Drop in flush mount register system with slidable damper

Abstract

A flush mount register system comprising: a register; a damper box; a substrate tray nested within the register via a plurality of struts; and having a continuous peripheral gap defined between the substrate tray and the register; a damper slidably mounted on the damper box for controlling air flow volume through the damper box into the continuous peripheral gap; wherein the substrate tray receives a substrate, and wherein the register rests on the damper box.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of U.S. patent application Ser. No. 17/127,602, filed Dec. 18, 2020, which is a continuation in part application of U.S. patent application Ser. No. 16/290,499, filed Mar. 1, 2019, which claims

FIELD

(1) The present disclosure relates to ventilation heating and cooling systems, and more particularly it relates to flush mount air registers.

BACKGROUND

(2) In forced air heating and cooling systems a fan associated with a furnace forces air through a duct system to distribute heated air or conditioned air throughout a building. Air emerges from outlets in the duct system, and the outlets generally include registers operable to regulate the flow of air. For registers which are situated in a floor, a grill is typically placed over the vent opening to provide an aesthetically pleasing appearance to the outlet. Normally, when installing flooring it is often desirable to include a register grill that also matches the flooring in order to give the flooring a homogeneous look. However, most flooring manufacturers and home improvement stores only have limited types and styles of register grills available, and end users are often forced to find the closest match to the flooring, and make do with their choice. Even if one was to consider custom manufacturing a limited number of register grills for a particular flooring, such a process may be impractical, expensive, and time consuming, as most manufacturers require minimum volumes to justify the cost of specialized tooling and dedicated personnel. In addition, given that there may be variations in flooring from batch to batch due to the nature of wood or tile, such matching may be difficult to achieve leading to inconsistencies in the overall flooring appearance.

SUMMARY

(3) In one aspect, a flush mount register system comprising: a register; a damper box; a substrate tray nested within the register via a plurality of struts; and having a continuous peripheral gap defined between the substrate tray and the register; a damper slidably mounted on the damper box for controlling air flow volume through the damper box into the continuous peripheral gap; wherein the substrate tray receives a substrate, and wherein the register rests on the damper box.

(4) In one aspect, a flush mount register system comprising: a register; a damper box comprising a top opening, a bottom opening and an apertured bottom wall adjacent to the bottom opening, and wherein the register rests on the damper box; a substrate tray nested within the register, and having a continuous peripheral gap defined between the substrate tray and the register, and wherein the substrate tray receives a substrate; a damper slidably mounted on the damper box for controlling air flow volume through the damper box into the continuous peripheral gap; an apertured grid removably attached to the bottom opening of the damper box, such that a cavity is defined between the apertured grid and the bottom opening; and an air filter retained within the cavity.

(5) Advantageously, the system comprises a damper box and substrate tray, which are both flush with the flooring, and can be used for flooring with various thicknesses e.g. 0.375 inches to 0.75 inches.

(6) In addition, the system can be customized for installation in flooring with various floor thicknesses, by simply adjusting the height of the flooring substrate frame such that flooring substrate frame, and tray with flooring substrate are flush with the rest of the flooring. Accordingly, there is no need multiple stock keeping units (SKUs) for each floor thickness or vent opening.

(7) Furthermore, the system allows any flooring system to be matched regardless of the flooring material. For example, the flooring material may be wood, laminate, carpet, engineered products, tile, carpet and vinyl.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 shows a perspective view of a flush mount register system;

- (2) FIG. 2 shows a top view of the flush mount register system;
- (3) FIG. 3 shows a sectional view of the flush mount register system;
- (4) FIG. 4 shows a perspective view of a register;
- (5) FIG. 5 shows a top perspective view of the register;
- (6) FIG. 6 shows a bottom perspective view of the register;
- (7) FIG. 7 shows another sectional view of the flush mount register system;
- (8) FIG. 8 shows a perspective view of a damper box;
- (9) FIG. 9 shows a perspective view of a damper;
- (10) FIG. 10 shows a sectional view of the flush mount register system, taken along line A-A in FIG. 2;
- (11) FIG. 11 shows a side view of the register;
- (12) FIG. 12 shows a perspective view of a grid; and
- (13) FIGS. 13-23 show different views of the flush mount register system, with various dimensions.

DETAILED DESCRIPTION

(14) The detailed description of exemplary embodiments herein makes reference to the accompanying block diagrams and schematic diagrams, which show the exemplary embodiment by way of illustration. While these exemplary embodiments are described in sufficient detail to enable those skilled in the art to practice the invention, it should be understood that other embodiments may be realized and that logical and mechanical changes may be made. Thus, the detailed description herein is presented for purposes of illustration only and not of limitation. For example, the steps recited in any of the method or process descriptions may be executed in any order and are not limited to the order presented.

(15) Moreover, it should be appreciated that the particular implementations shown and described herein are illustrative of the invention and its best mode and are not intended to otherwise limit the scope of the present invention in any way. Indeed, for the sake of brevity, certain sub-components of the individual operating components.

(16) Looking at FIGS. 1-3 there is shown a flush mount register system **10** for installation in an opening of a surface, such as a floor, ceiling, or wall. In one example, the surface may be covered by a material, such as wood, laminate, composite, plastic, tile, or drywall. The opening is in fluid communication with duct coupled to a heating, ventilation, and air conditioning (HVAC) system. Generally, flush mount register system **10** comprises register **12**, damper box **14** and a substrate tray **16** which receives a substrate **18**, such as a flooring substrate. Preferably, the substrate **18** that is identical to the surface in which the flush mount register system is mounted, which leads to a more desirable aesthetic look.

(17) Now referring to FIGS. 4-6, register **12** comprises first pair of longitudinally extending sidewalls **20**, **22** and first pair of laterally extending sidewalls **24**, **26**, which are joined to form a rectangularly shaped structure with top opening **27a** and bottom opening **27b**. Longitudinally extending walls **20**, **22** and laterally extending sidewalls **24**, **26** comprise rectangular flange **28** extending outwardly and orthogonally from top edges **30**, **31** of longitudinally extending walls **20**, **22**, and extending outwardly and orthogonally from top edges **32**, **33** of first pair of laterally extending sidewalls **24**, **26**. Bottom edges **34**, **35** of first pair of longitudinally extending sidewalls **20**, **22** and bottom edges **36**, **37** of first pair of laterally extending sidewalls **24**, **26**, define bottom opening **27b**. The first pair of laterally extending sidewalls **24**, **26** comprises two portions **24a**, **24b** and **26a**, **26b**, respectively, with a shoulder **38** defined between two portions **24a**, **24b** and **26a**, **26b**, respectively. Substrate tray **16** is disposed between top opening **27a** and bottom opening **27b**, and is dimensioned such that a continuous peripheral gap **39** is defined between the substrate tray **16** walls and longitudinally extending walls **20**, **22** and laterally extending sidewalls **24**, **26** of register **12**. Portions **24b**, **26b** of first pair of laterally extending sidewalls **24**, **26**, respectively, include one or more crevices **41** which engage tabs **82** on the damper box **14**, as will be described below with

reference to FIG. 8.

(18) As can be seen in FIGS. 5 and 6, substrate tray **16** comprises second longitudinally extending walls **40**, **42** and laterally extending walls **44**, **46**, joined together to form a rectangularly shaped structure, with top opening **48** and base **50**. Struts **52** extend between the second longitudinally extending walls **40**, **42** and the first pair of longitudinally extending sidewalls **20**, **22** of the register and maintain the substrate tray **16** within the register **12** with the gap **39** therebetween. Air from the duct is forced through the gap **39** into the space. The base **50** of substrate tray **16** comprises air deflector **60** for directing air towards gap **39**, thereby increasing air flow through gap **39**.

(19) Looking at FIG. 7, flooring substrate **18** is dimensioned to fit within substrate tray **16**, such that substrate tray **16** is flush with flooring **12**. The depth of substrate tray **16** may be adjusted, or may be varied to allow for full customization of flush mount register system **10** to match the various thicknesses of flooring **12**. Accordingly, flush mount register system **10** can be easily installed, without having to level any other component of flush mount register system **10**.

(20) Now referring to FIG. 8, damper box **14** comprises third pair of longitudinally extending sidewalls **70**, **72** and third pair of laterally extending sidewalls **74**, **76**, which are joined to form a rectangularly shaped structure with top opening **77a** and bottom opening **77b**. Located between the top opening **77a** and the bottom opening **77b** is a bottom wall **78** with a plurality of apertures **80** that are spaced apart by a predefined distance to allow for air to pass through from the duct into the damper box **14**. Third pair of longitudinally extending sidewalls **70**, **72** and third pair of laterally extending sidewalls **74**, **76** comprise an edge **81**. Third pair of laterally extending sidewalls **74**, **76** further comprise one or more tabs **82**. The register **12** is placed on the damper box **14** such that the edges **81** rest on shoulder **38**, and the tabs **82** are caused to engage one or more crevices **41** of the register **12** to removably attach the damper box **14** to the register **12**.

(21) A damper **90** is slidably mounted within the damper box **14**, and rests on the apertured bottom wall **78** adjacent to the damper box bottom opening **77b**, for controlling air flow into the damper box **14**. As can be seen in FIG. 9, the damper **90** comprises a sliding plate **92** with a plurality of apertures **94** that are spaced apart by a predefined distance, and is configured to move in one direction within the damper box **14** such that the plurality of apertures **94** are aligned, partially aligned, or misaligned with the plurality of apertures **80** of the damper box **14** to allow regulate air flow into the damper **14**. The damper **90** comprises a finger **96** projecting from a top surface **98**, the finger **96** being located between wall **46** of the substrate tray **16** and wall **76** of the damper box **14**. The finger **96** is dimensioned such that it is graspable by a user, such that the user can apply a force to cause the damper **14** to linearly slide relative to the bottom wall **78** of the damper box **14**.

Accordingly, as the sliding plate **92** slides linearly back and forth relative to the apertured bottom wall **78**, the apertures **80**, **94** align to form airflow passages of a variable size, thereby regulating the airflow into the damper box **14**. As such, when the apertures **80**, **94** are fully aligned then maximum air flow into the damper box **14** may be achieved, whereas when the apertures **80**, **94** are fully misaligned then air flow into the damper box **14** is substantially impeded or blocked. In one example, the apertures **80** in the bottom **78** and the apertures **94** of the sliding plate **92** are substantially rectangular-shaped, however the apertures **94** may be formed to have shapes other than rectangular.

(22) In one example, apertured grid **99** is removably attached to the bottom opening **77b** such that a cavity **102** is defined between the grid **99** and the apertured bottom wall **78**, as can be seen in FIGS. **10-12**. The grid **99** comprises a plurality of apertures **100**, and may be configured to frictionally fit within the bottom opening **77b**. The cavity **102** may receive an air filter **103** which is retained therein by the grid **99**. In one example, grid **99** is removably attached to the bottom opening **77b**, as such, the air filter **103** can be replaced periodically as needed.

(23) As can be seen in FIGS. **13-23**, the register system **10** may include components **12**, **14**, **16**, **18**, **19**, **90**, **99** with differing dimensions.

(24) In addition, damper box **14** and substrate tray **16** are dimensioned to fit inside industry

standard vent openings. Accordingly, when flush mount register system **10** is fully assembled, substrate tray **16** is capable of supporting similar forces imposed on flooring **12**, such as those from furniture, objects, or people.

(25) In another implementation, damper box **14** includes a single damper or any number of dampers, depending on the size of the opening, application, desired results, and user preferences.

(26) In another implementation, while the system **10** is primarily meant to be installed at the same time as the flooring material **18**, it can be modified to allow use of leftover flooring such that the finished product matches the existing flooring material **18**.

(27) In another implementation, substrate tray **16** is removably secured to damper box **14**.

(28) Those skilled in the art will recognize that the dimensions shown in the drawings are exemplary in nature only, as these dimensions are applicable in one exemplary implementation e.g. for 4"×10" vent openings, and therefore can vary depending on the requirements for a particular application, desired results, and user preferences. For example, flush mount register system **10** may be dimensioned to fit any opening, accordingly exemplary sizes of flush mount register system **10** may be 2"×12" and 4"×12".

(29) In another implementation, the system **10** is installed in a ceiling or a wall. Accordingly, substrate tray **16** may house ceiling tile or drywall.

(30) In another implementation, each of the pair of second laterally extending walls **44**, **46** of substrate tray **16** comprise at least one strut extending outwardly, wherein one end of the struts **52**, **54** is secured to the first pair of laterally extending sidewalls **24**, **26** of damper box **14**.

(31) The preceding detailed description of exemplary embodiments of the invention makes reference to the accompanying drawings, which show the exemplary embodiment by way of illustration. While these exemplary embodiments are described in sufficient detail to enable those skilled in the art to practice the invention, it should be understood that other embodiments may be realized and that logical and mechanical changes may be made. For example, the steps recited in any of the method or process claims may be executed in any order and are not limited to the order presented. Further, the present invention may be practiced using one or more servers, as necessary. Thus, the preceding detailed description is presented for purposes of illustration only and not of limitation, and the scope of the invention is defined by the preceding description, and with respect to the attached claims.

Claims

1. A flush mount register system comprising: a register; a damper box; a substrate tray nested within the register via a plurality of struts; and having a continuous peripheral gap defined between the substrate tray and the register; a damper slidably mounted on the damper box for controlling air flow volume through the damper box into the continuous peripheral gap; wherein the substrate tray receives a substrate, and wherein the register rests on the damper box.
2. The flush mount register system of claim 1, wherein the register comprises a first pair of longitudinally extending sidewalls and a first pair of laterally extending sidewalls which are joined to form a rectangularly shaped structure with a register top opening and a register bottom opening, wherein the first pair of laterally extending sidewalls comprises two portions with a shoulder defined therebetween.
3. The flush mount register system of claim 2, wherein the substrate tray comprises a second pair of longitudinally extending sidewalls and a second pair of laterally extending sidewalls which are joined to form a rectangularly shaped structure with a base and a substrate tray opening, and wherein the plurality of struts are secured the first pair of longitudinally extending sidewalls at one end and to the second pair of longitudinally extending sidewalls.
4. The flush mount register system of claim 3, wherein the substrate is contained within the second pair of longitudinally extending sidewalls and the second pair of laterally extending sidewalls.

5. The flush mount register system of claim 4, wherein the damper box comprises a damper box top opening and a damper box bottom opening with an apertured bottom wall.
 6. The flush mount register system of claim 5, wherein the damper comprises a plate with a first of plurality of apertures.
 7. The flush mount register system of claim 6, wherein the apertured bottom wall comprises a second plurality of apertures.
 8. The flush mount register system of claim 7, wherein the plate is slidable over the apertured bottom wall such that the first of plurality of apertures are at least aligned, partially aligned, or misaligned with the second plurality of apertures, thereby regulating the air flow volume into the damper box.
 9. The flush mount register system of claim 3, wherein the base of the substrate tray comprises an air deflector for directing air towards the continuous peripheral gap, thereby increasing the air flow volume through the continuous peripheral gap.
 10. The flush mount register system of claim 8, wherein the damper box and the substrate tray are coplanar, such that the substrate is flush with a surrounding flooring.
 11. The flush mount register system of claim 8, further comprising an apertured grid is removably attached to the bottom opening such that a cavity is defined between the apertured grid and the damper box bottom opening.
 12. The flush mount register system of claim 11, wherein an air filter is retained within the cavity.
 13. The flush mount register system of claim 8, wherein the damper comprises a finger projecting from a top surface of the damper, the finger extending through the continuous peripheral gap, whereby a force is applied to the finger to cause the damper to linearly slide relative to the apertured bottom wall of the damper box.
 14. A flush mount register system comprising: a register; a damper box comprising a top opening, a bottom opening and an apertured bottom wall adjacent to the bottom opening, and wherein the register rests on the damper box; a substrate tray nested within the register, and having a continuous peripheral gap defined between the substrate tray and the register, and wherein the substrate tray receives a substrate; a damper slidably mounted on the damper box for controlling air flow volume through the damper box into the continuous peripheral gap; an apertured grid removably attached to the bottom opening of the damper box, such that a cavity is defined between the apertured grid and the bottom opening; and an air filter retained within the cavity.
 15. The flush mount register system of claim 14, wherein the damper comprises a plate with a first of plurality of apertures.
 16. The flush mount register system of claim 15, wherein the apertured bottom wall comprises a second plurality of apertures.
 17. The flush mount register system of claim 16, wherein the plate is slidable over the apertured bottom wall such that the first of plurality of apertures are at least aligned, partially aligned, or misaligned with the second plurality of apertures, thereby regulating the air flow volume into the damper box.
 18. The flush mount register system of claim 14, wherein the base of the substrate tray comprises an air deflector for directing air towards the continuous peripheral gap, thereby increasing the air flow volume through the continuous peripheral gap.
 19. The flush mount register system of claim 14, wherein the damper comprises a finger projecting from a top surface of the damper, the finger extending through the continuous peripheral gap, whereby a force is applied to the finger to cause the damper to linearly slide relative to the apertured bottom wall of the damper box.
 20. The flush mount register system of claim 14, wherein the substrate tray secured to interior walls of the register via at least one strut.
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